Fm16

Con McIlveen Numerical Methods 2

Review

1 $y' = -3x^2 + 6x$

$y'' = -6x + 6$

@ point of inflection, $y'' = 0$

$-6x = -6$
$x = 1$

$y(1) = -1 + 3 - 1$
$= 1$

$\therefore$ point of inflection @ $(1,1)$

2 $s = ka^t$

$\log s = \log(ka^t)$

$= \log k + \log a^t$
$= \log k + t \log a$

graph is $\log s$ against $\log a$

$\log k = 0.3 \quad \rightarrow k = 10^{\frac{3}{10}}$
$t = 0.8 \qquad = 2.0$

4 $\int (2x+1)\ln x \, dx = (x^2 + x)\ln x - \int \frac{x^2+x}{x} dx$

$= (x^2 + x)\ln x - \frac{1}{2}x^2 - x \quad (c)$

5 $\int \frac{x-2}{x^2(x+1)} dx$

$\frac{A}{x} + \frac{B}{x+1} + \frac{C}{x^2} \qquad = \frac{x-2}{x^2(x+1)}$

$Ax(x+1) + B(x^2) + C(x+1) = x - 2$

$C = -2$
$B = -3$
$A = 3 \quad Ax^2 + Bx^2 = 0$

$\int 3x^{-1} - 3(x+1)^{-1} - 2x^{-2} \, dx$

$= 3\ln|x| - 3\ln|x+1| + 2x^{-1} = 3\ln\frac{x}{x+1} + \frac{2}{x} \quad c$

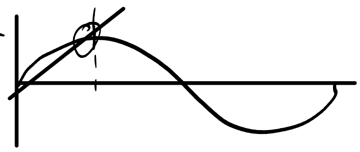
1 $e^x - 3x + 0.2$: @ 1.34, $= -9.56\times10^{-4} < 0$
@ 1.35, $= 7.43 > 0$
Sign change $\therefore$ root

2 $x^3 - 7x - 12$: @ 3.2665 $= -0.01 < 0$
@ 3.2675 $= 0.01 > 0$
Sign change $\therefore$ root

3 $x^3 - x - 1 = 0$
$x^3 = x + 1$
$x = (x+1)^{\frac{1}{3}}$

At $x_0 = 1$:
$x_0 = 1$
$x_1 = 1.2599$
$x_2 = 1.3123$
$x_3 = 1.3224$
$x_4 = 1.3243$
$x_5 = 1.3246$
$x_6 = 1.3247$ $\therefore$ root $= 1.325$ (3dp)

(ii) $x^3 - x - 1$ @ 1.3245 $= -9.243$
@ 1.3255 $= 3.338$

4


ii) Let $x_0 = \frac{\pi}{3}$

Then $x_1 = 1.3660$
$x_2 = 1.4791$
$x_3 = 1.4958$
$x_4 = 1.4972 \approx 1.497$
$x_5 = 1.4973 \approx 1.497$

iii) at $x = 1.4965$,
$2\sin x - 2x + 1 = 1.4826 > 0$
at $x = 1.4975$, $\uparrow = -4\times10^{-4} < 0$
sign change $\therefore$ root present in interval.

5 $y = 2x^3 + 4x - 5$
$y' = 6x^2 + 4$
$6x^2 \geq 0$
$4 > 0$
$\therefore 6x^2 + 4 > 0$
so gradient always positive.

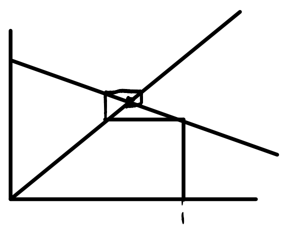
If gradient always positive, range of y is $-\infty$ to ∞

b) At $x=0$, $y=-5$ <0
At $x=1$, $y=6-5=1$ >0
Sign change $\therefore$ root

c) $2x^3+4x-5=0$

$x(2x^2+4)-5=0$
$x=\frac{5}{2x^2+4}$

d) $x_1=\frac{5}{6}=0.833$
$x_2=\frac{90}{97}=0.928$



At 0.8925, $y=-8.117\times10^{-3}$ <0
At 0.8935, $y=6.38\times10^{-4}$ >0
Sign change $\therefore$ root

6 iv) 1 and 2. Let $f(x)$ be the originally-defined fn.
Then $f'(x)=5x^4-1$
Let $x_0=1.5$.
Then $x_1=x_0-\frac{f(x_0)}{f'(x_0)}$
$x_5=1.673$
$\therefore$ root @ 1.673

v) 0 and 1. $f'(x)=9x^2+4x$

Let $x_0=0.5$.
Then $x_1=1$
$x_2=0.8462$
$x_3=0.8208$
$x_4=0.8201$
$x_5=0.8201$
$\therefore$ root at 0.820

7 total area $=\frac{1}{2}r^2\theta=80\theta$
triangle area $=\frac{1}{2}ab\sin\theta$
$=50\sin\theta$. half area.
$50\sin\theta=25\theta$
$2\sin\theta-\theta=0$

b) $\frac{d}{d\theta} = 2\cos\theta - 1$

$$\theta_{n+1} = \theta_n - \frac{2\sin\theta - \theta}{2\cos\theta - 1}$$

$\theta_1 = 1.9010$
$\theta_2 = 1.8965$
$\theta_3 = 1.8954$
$\theta_4 = 1.8954$
So $\theta = 1.8954$

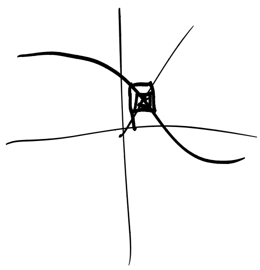
Exam Qs

1a) $\sin x = -\frac{1}{2}x + \frac{1}{3}\pi$

$\sin x + \frac{1}{2}x - \frac{1}{3}\pi = 0$

At $x = 0$, $f = -\frac{1}{3}\pi < 0$
At $x = 1$, $f = 1.01 > 0$
Sign change $\therefore$ root

b) $x_0 = 0.5$
$x_1 = 0.613$
0.618
0.635
0.630
0.632
0.631
0.631



2 a) $f(x) = x^3 - x^2 - 5x + 10$
$f'(x) = 3x^2 - 2x - 5$

$$x_{n+1} = x_n - \frac{f(x)}{f'(x)}$$
$$= x_n - \frac{x_n^3 - x_n^2 - 5x_n + 10}{3x_n^2 - 2x_n - 5}$$

$$= \frac{2x_n^3 - x_n^2 - 10}{3x_n^2 - 2x_n - 5}$$

b) $x_1 = -3$
$x_2 = -2.607$
$x_3 = -2.535$
$x_4 = -2.533$

c $f(-2.5329) = 6.6\times10^{-3} > 0$
$f(-2.5333) = -0.01 < 0$

Sign change $\therefore$ root

d) -1 is too close to a stationary point
$\therefore$ would not converge.